KAIZEN MATHS

Lesson starter

Trigonometric Differentiation

Advanced Trig Compositions

Name _____

S1

Differentiate with respect to x.

$y = (\cos(-3x^2 - 8x - 4))^3$

S2

Differentiate with respect to x.

$y = (\sec(3x + 8))^2 \sin(-9x - 7)$

S3

Differentiate with respect to x.

$y = (\cot(-3x^2 + 8x - 7))^4$

S4

Differentiate with respect to x.

$y = (\tan(-4x + 3))^3 \sec(-5x + 8)$

Show your working

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ALG-C0E7-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Trigonometric Differentiation

Advanced Trig Compositions

Name _____

E1

Differentiate with respect to x.

$y = (\sin(-3x^2 - 4x - 9))^3$

E2

Differentiate with respect to x.

$y = (\sin(-4x - 1))^3 \cot(8x + 7)$

E3

Differentiate with respect to x.

$y = (\cot(-3x^2 + 2x + 6))^4$

KAIZEN MATHS

Exit ticket

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Advanced Trig Compositions

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Differentiate with respect to x.

$y = (\sin(-3x^2 - 4x - 9))^3$

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Differentiate with respect to x.

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ALG-C0E7-L3 · E

Show your working

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Exit ticket

Trigonometric Differentiation

Advanced Trig Compositions

Name _____

E1

Differentiate with respect to x.

$y = (\sin(-3x^2 - 4x - 9))^3$

E2

Differentiate with respect to x.

$y = (\sin(-4x - 1))^3 \cot(8x + 7)$

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Differentiate with respect to x.

$y = (\cot(-3x^2 + 2x + 6))^4$

ALG-C0E7-L3 · E

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Advanced Trig Compositions

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E1

Differentiate with respect to x.

$y = (\sin(-3x^2 - 4x - 9))^3$

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Differentiate with respect to x.

$y = (\sin(-4x - 1))^3 \cot(8x + 7)$

E3

Differentiate with respect to x.

$y = (\cot(-3x^2 + 2x + 6))^4$

ALG-C0E7-L3 · E

Show your working

Teacher answers & working

Trigonometric Differentiation · Advanced Trig Compositions

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S1

Differentiate with respect to x.

$y = (\cos(-3x^2 - 8x - 4))^3$

Answer:

$\frac{dy}{dx} = -3\,(\cos(-3x^2 - 8x - 4))^2\,\sin(-3x^2 - 8x - 4)\,(-6x - 8)$

Working:

Work from the outside in: power, then trig function, then polynomial inner function.

Outer power derivative gives $3(\cos(-3x^2 - 8x - 4))^2$.

Trig derivative gives $-\sin(-3x^2 - 8x - 4)$.

Inner derivative gives $-6x - 8$.

Multiply the three derivative parts together.

$\frac{dy}{dx} = -3\,(\cos(-3x^2 - 8x - 4))^2\,\sin(-3x^2 - 8x - 4)\,(-6x - 8)$

S2

Differentiate with respect to x.

$y = (\sec(3x + 8))^2\,\sin(-9x - 7)$

Answer:

$\frac{dy}{dx} = 3\,(\sec(3x + 8))^2[2\,\tan(3x + 8)\,\sin(-9x - 7) - 3\,\cos(-9x - 7)]$

Working:

This is a product, and the first factor also needs the chain rule.

Let $f = (\sec(3x + 8))^2$ and $g = \sin(-9x - 7)$.

$f' = 2(\sec(3x + 8))^{2-1}(\sec(3x + 8)\tan(3x + 8))(3)$

$f' = 6\,(\sec(3x + 8))^1\,\sec(3x + 8)\tan(3x + 8)$

$g' = (\cos(-9x - 7))(-9)$

$g' = -9\,\cos(-9x - 7)$

Apply product rule: $y' = f'g + fg'$.

$y' = (6\,(\sec(3x + 8))^1\,\sec(3x + 8)\tan(3x + 8))(\sin(-9x - 7)) + ((\sec(3x + 8))^2)(-9\,\cos(-9x - 7))$

$y' = 6\,(\sec(3x + 8))^1\,\sec(3x + 8)\tan(3x + 8)\,\sin(-9x - 7) - 9\,(\sec(3x + 8))^2\,\cos(-9x - 7)$

Factor out the common factor $3\,(\sec(3x + 8))^2$.

Factorised final form: $\frac{dy}{dx} = 3\,(\sec(3x + 8))^2[2\,\tan(3x + 8)\,\sin(-9x - 7) - 3\,\cos(-9x - 7)]$

Teacher answers & working

Trigonometric Differentiation · Advanced Trig Compositions

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S3

Differentiate with respect to x.

$y = (\cot(-3x^2 + 8x - 7))^4$

Answer:

$\frac{dy}{dx} = -4\,(\cot(-3x^2 + 8x - 7))^3\ \csc^2(-3x^2 + 8x - 7)\,(-6x + 8)$

Working:

Work from the outside in: power, then trig function, then polynomial inner function.

Outer power derivative gives $4(\cot(-3x^2 + 8x - 7))^3$.

Trig derivative gives $-\csc^2(-3x^2 + 8x - 7)$.

Inner derivative gives $-6x + 8$.

Multiply the three derivative parts together.

$\frac{dy}{dx} = -4\,(\cot(-3x^2 + 8x - 7))^3\ \csc^2(-3x^2 + 8x - 7)\,(-6x + 8)$

S4

Differentiate with respect to x.

$y = (\tan(-4x + 3))^3\ \sec(-5x + 8)$

Answer:

$\frac{dy}{dx} = -(\tan(-4x + 3))^2\ \sec(-5x + 8)[12\,(\sec(-4x + 3))^2 + 5\ \tan(-4x + 3)\ \tan(-5x + 8)]$

Working:

This is a product, and the first factor also needs the chain rule.

Let $f = (\tan(-4x + 3))^3$ and $g = \sec(-5x + 8)$.

$f' = 3(\tan(-4x + 3))^{3-1}(\sec^2(-4x + 3))(-4)$

$f' = -12\,(\tan(-4x + 3))^2\ \sec^2(-4x + 3)$

$g' = (\sec(-5x + 8)\tan(-5x + 8))(-5)$

$g' = -5\ \sec(-5x + 8)\tan(-5x + 8)$

Apply product rule: $y' = f'g + fg'$.

$y' = (-12\,(\tan(-4x + 3))^2\ \sec^2(-4x + 3))(\sec(-5x + 8)) + ((\tan(-4x + 3))^3)(-5\ \sec(-5x + 8)\tan(-5x + 8))$

$y' = -12\,(\tan(-4x + 3))^2\ \sec^2(-4x + 3)\ \sec(-5x + 8) - 5\,(\tan(-4x + 3))^3\ \sec(-5x + 8)\tan(-5x + 8)$

Factor out the common factor $-(\tan(-4x + 3))^2\ \sec(-5x + 8)$.

Factorised final form: $\frac{dy}{dx} = -(\tan(-4x + 3))^2\ \sec(-5x + 8)[12\,(\sec(-4x + 3))^2 + 5\ \tan(-4x + 3)\ \tan(-5x + 8)]$

Teacher answers & working

Trigonometric Differentiation · Advanced Trig Compositions

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E1

Differentiate with respect to x.

$y = (\sin(-3x^2 - 4x - 9))^3$

Answer:

$\frac{dy}{dx} = 3\,(\sin(-3x^2 - 4x - 9))^2\,\cos(-3x^2 - 4x - 9)\,(-6x - 4)$

Working:

Work from the outside in: power, then trig function, then polynomial inner function.

Outer power derivative gives $3(\sin(-3x^2 - 4x - 9))^2$.

Trig derivative gives $\cos(-3x^2 - 4x - 9)$.

Inner derivative gives $-6x - 4$.

Multiply the three derivative parts together.

$\frac{dy}{dx} = 3\,(\sin(-3x^2 - 4x - 9))^2\,\cos(-3x^2 - 4x - 9)\,(-6x - 4)$

E2

Differentiate with respect to x.

$y = (\sin(-4x - 1))^3\,\cot(8x + 7)$

Answer:

$\frac{dy}{dx} = -4\,(\sin(-4x - 1))^2[3\,\cos(-4x - 1)\,\cot(8x + 7) + 2\,\sin(-4x - 1)\,(\csc(8x + 7))^2]$

Working:

This is a product, and the first factor also needs the chain rule.

Let $f = (\sin(-4x - 1))^3$ and $g = \cot(8x + 7)$.

$f' = 3(\sin(-4x - 1))^{3-1}(\cos(-4x - 1))(-4)$

$f' = -12\,(\sin(-4x - 1))^2\,\cos(-4x - 1)$

$g' = (-\csc^2(8x + 7))(8)$

$g' = -8\,\csc^2(8x + 7)$

Apply product rule: $y' = f'g + fg'$.

$y' = (-12\,(\sin(-4x - 1))^2\,\cos(-4x - 1))(\cot(8x + 7)) + ((\sin(-4x - 1))^3)(-8\,\csc^2(8x + 7))$

$y' = -12\,(\sin(-4x - 1))^2\,\cos(-4x - 1)\,\cot(8x + 7) - 8\,(\sin(-4x - 1))^3\,\csc^2(8x + 7)$

Factor out the common factor $-4\,(\sin(-4x - 1))^2$.

Factorised final form: $\frac{dy}{dx} = -4\,(\sin(-4x - 1))^2[3\,\cos(-4x - 1)\,\cot(8x + 7) + 2\,\sin(-4x - 1)\,(\csc(8x + 7))^2]$

Teacher answers & working

Trigonometric Differentiation · Advanced Trig Compositions

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E3

Differentiate with respect to x.

$y = (\cot(-3x^2 + 2x + 6))^4$

Answer:

$\frac{dy}{dx} = -4\left(\cot(-3x^2 + 2x + 6)\right)^3 \csc^2(-3x^2 + 2x + 6)\left(-6x + 2\right)$

Working:

Work from the outside in: power, then trig function, then polynomial inner function.

Outer power derivative gives $4(\cot(-3x^2 + 2x + 6))^3$.

Trig derivative gives $-\csc^2(-3x^2 + 2x + 6)$.

Inner derivative gives $-6x + 2$.

Multiply the three derivative parts together.

$\frac{dy}{dx} = -4\left(\cot(-3x^2 + 2x + 6)\right)^3 \csc^2(-3x^2 + 2x + 6)\left(-6x + 2\right)$